

HealthConnect Week 6 Cross-Track Integration Evidence

Data Analytics ↔ Data Science

❖ Track collaborated with

Data Science.

❖ Project dependency

The Data Science models require business-relevant feature and segment evidence, while the Data Analytics dashboard benefits from model outputs that show which factors contribute to predictive risk and how well probability scores separate no-shows from attendance.

❖ Information/output received

- Random Forest candidate feature importance and threshold/error-analysis summary.
- Signed Logistic Regression coefficient file.
- Logistic Regression test-set scoring file containing actual outcome, predicted probability, and predicted class.
- Initial guidance that final operational Low/Medium/High thresholds require stakeholder cost/capacity input.

❖ Information/output provided

- Validated no-show rates by lead time, previous no-show history, reminders, distance, and appointment type.
- Advanced interaction findings including lead time × previous history, lead time × reminder, lead time × distance and type × history.
- Statistical validation showing lead time as the strongest segment-level association.
- Business interpretation and recommended priority segments for feature refinement/error analysis.

❖ Integration activity completed

Data Science outputs were used to prioritise Week 6 Analytics investigations. Analytics then tested the model-related factors in the raw appointment data and translated model scoring behaviour into stakeholder-friendly decision-support outputs.

❖ What changed as a result

The Week 6 dashboard/report now focuses on validated high-impact segments rather than giving equal weight to every variable. Lead time and previous no-show history became the main decision-support story. The integration also revealed an interface problem: the scoring export does not retain the true HealthConnect appointment ID, so row-level operational model error analysis is deferred to Week 7.

❖ Evidence

Evidence item	Result
Refined lead-time analysis	41–60 days = 68.5% no-show rate
Interaction analysis	41–60 days + 2+ prior no-shows = 83.3%
Reminder interaction	Reminder-associated rate is lower within every lead-time band
Distance interaction	20+ km remains elevated, including 72.9% at 41–60 days
Model scoring validation	ROC-AUC $\approx$ 68.9%; lowest score decile $\approx$ 16.5% actual no-shows, highest $\approx$ 77.3%
Integration issue log	Scoring integer ID cannot be joined to original HC appointment ID
Dashboard artefact	Week 6 dashboard input workbook and two-page redesign plan

❖ Integration conclusion

The collaboration produced a measurable change in the Analytics output: Week 6 moved from descriptive EDA to advanced segmentation, model-informed validation, and a documented testing plan for the next project stage.